

STEFANO PAGGI

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EXPERIENCE

University of Toulouse

2023-2026

PhD researcher

Toulouse, France

- **Technical project ownership:** Developed a high-performance Python/FORTRAN framework for simulating electron excitation spectra using advanced many-body methods.
- **Complex problem solving:** Engineered numerical solutions for extreme-scale linear algebra problems involving matrices with up to 10^{11} elements, reducing problems to their essential linear subspaces.
- **Performance Optimization:** Optimized matrix construction and diagonalization pipelines, achieving significant performance gains in large-scale simulations.
- **High Performance Computing (HPC):** Deployed and executed workloads on HPC systems using slurm.
- **Quality and validation:** Developed numerically stable algorithms, benchmarked results against the literature, and validated implementations through independent computational approaches.
- **Continuous improvement:** Performed quantitative analysis of computational results and used findings to iteratively optimize implementations.
- **Cross-functional / international collaboration:** Collaborated with a research team in Amsterdam to extend computational methods from molecular systems to solids.
- **Technical communication:** Maintained code repositories and online documentation and communicated research results through international presentations.

International Summer Course at University of Heidelberg

August 2023

Tutor (singled out for exceptional motivation)

Heidelberg, Germany

- Tutored a class of international students through day-to-day activities.
- Organized extracurricular activities for international students.
- Collaborated on concerts and the final celebration as master of ceremony.
- Guide of the castle of Schwetzingen.

BLS English/Fondazione Vasillij Grossman

June 2023

Assistant teacher at student school trip

Bury St. Edmunds, UK

- Assisted the principal teacher on a school trip from Milan, Italy to England.
- Oversaw day-to-day activities of about 30 Italian students.

EDUCATION

Heidelberg University

2020 - 2023

M.Sc., Condensed Matter Physics, 2023 (GPA: 3.8/4.0 (German scale: 1.2, High Distinction))

Heidelberg, Germany

- Applied the Quanta package using Lua scripting and Wolfram Mathematica to model cerium-silicon complex at nonzero temperature (Boltzmann statistics). Results were checked against experiment.

Università degli Studi di Roma Tor Vergata

2021 - 2022

ERASMUS+ exchange program

Rome, Italy

Heidelberg University

2016 - 2020

B.Sc., Physics, 2020 (German scale: 2.5, Good)

Heidelberg, Germany

- Collaborated within an experimental physics group on the modeling and analysis of a ^3He atomic beam spectrometer.
- Computed beam trajectories and scattering cross sections, supporting the design and interpretation of experiments.
- Developed tools for wavelet scattering representations for signal characterization.
- **Relevant Coursework:** Optics, Electromagnetism, Linear Algebra, Scientific Computing, Numerical Methods.

LEADERSHIP & VOLUNTEERING

Leader of English student parish

2024 – Present

Student parish of Toulouse

Toulouse, France

- Founded & led an international student community, coordinating activities and collaborating with members from different cultural and national backgrounds.
- Organized and communicated across a diverse group of students and stakeholders.
- Developed experience in leadership, coordination, relationship building, and working across cultural differences.

SKILLS & OUTREACH

Technical: Python (NumPy, SciPy, PySCF), Wolfram Mathematica, Lua, Java, FORTRAN, Git, emacs, L^AT_EX, Linux/Shell, High Performance Computing (Slurm), MPI.

Languages: German (native), Italian (fluent), English (fluent), French (beginner).

Publications: 4 publications with 2 first-author publications in high impact journals, 6 talks and presentations given at international venues.